Lesson 05 Demo 03

Creating a React App with the useReducer Hook

Objective: To create a React app using Vite and implement the useReducer Hook for efficient state management in a counter application

Tools required: Visual Studio Code and Node.js

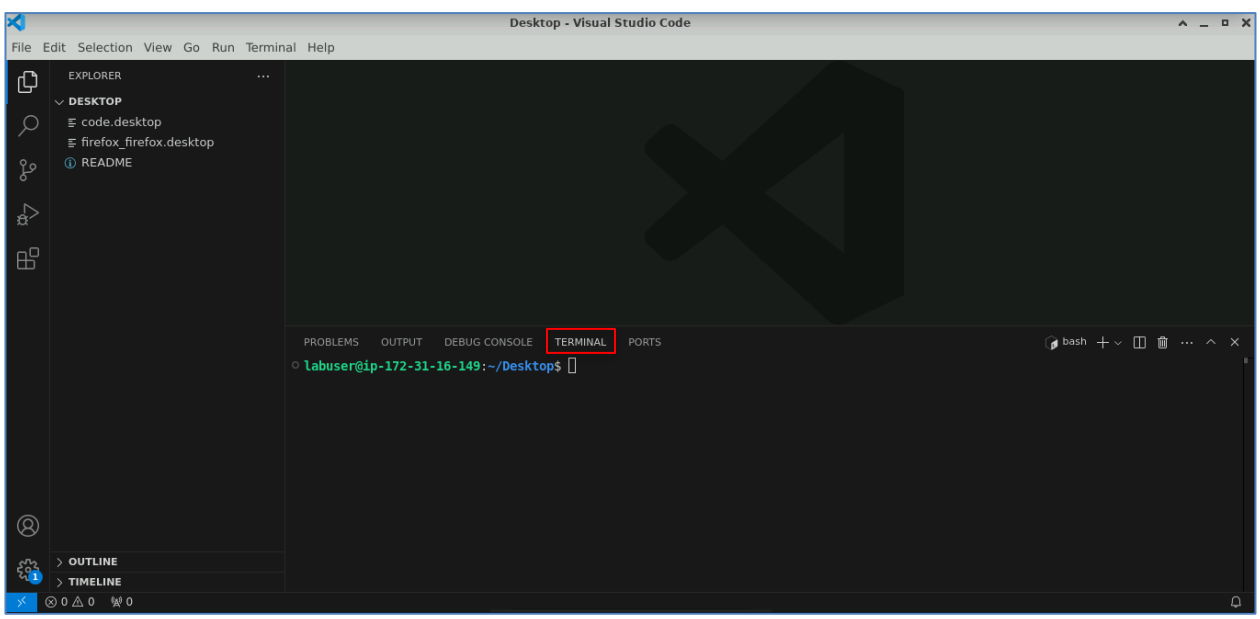
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Modify the App.jsx file
3. Run and verify the application

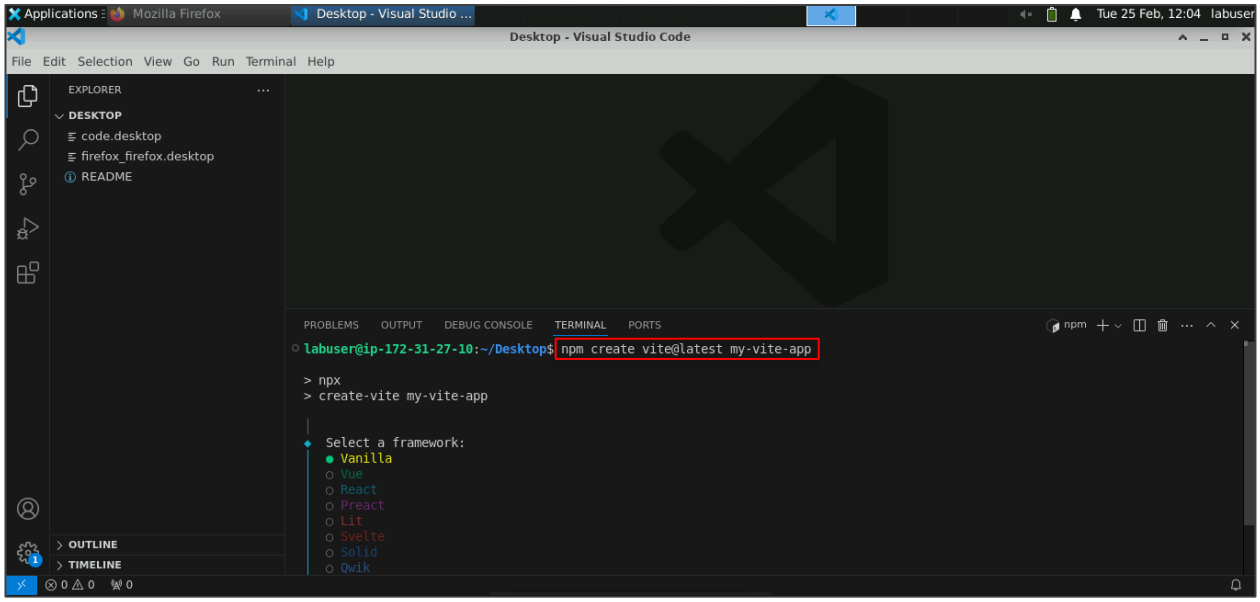
Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code (VS Code) and launch the terminal

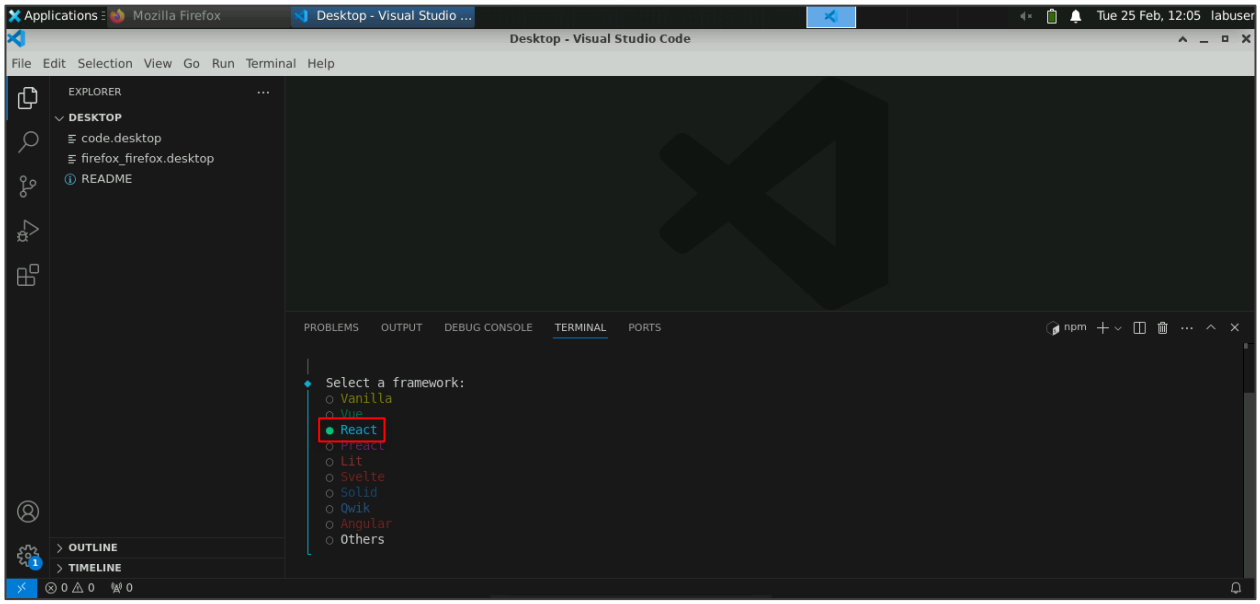


1.2 Run the following command to create a new React project using Vite:

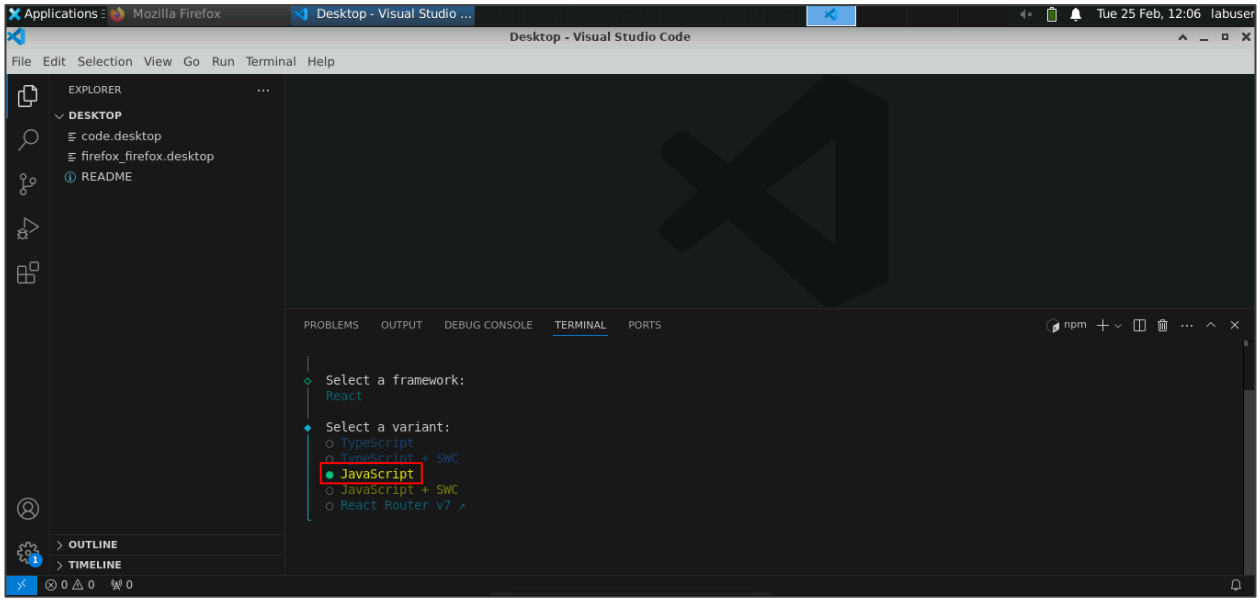
npm create vite@latest my-vite-app



1.3 Select **React** as the project framework and press enter

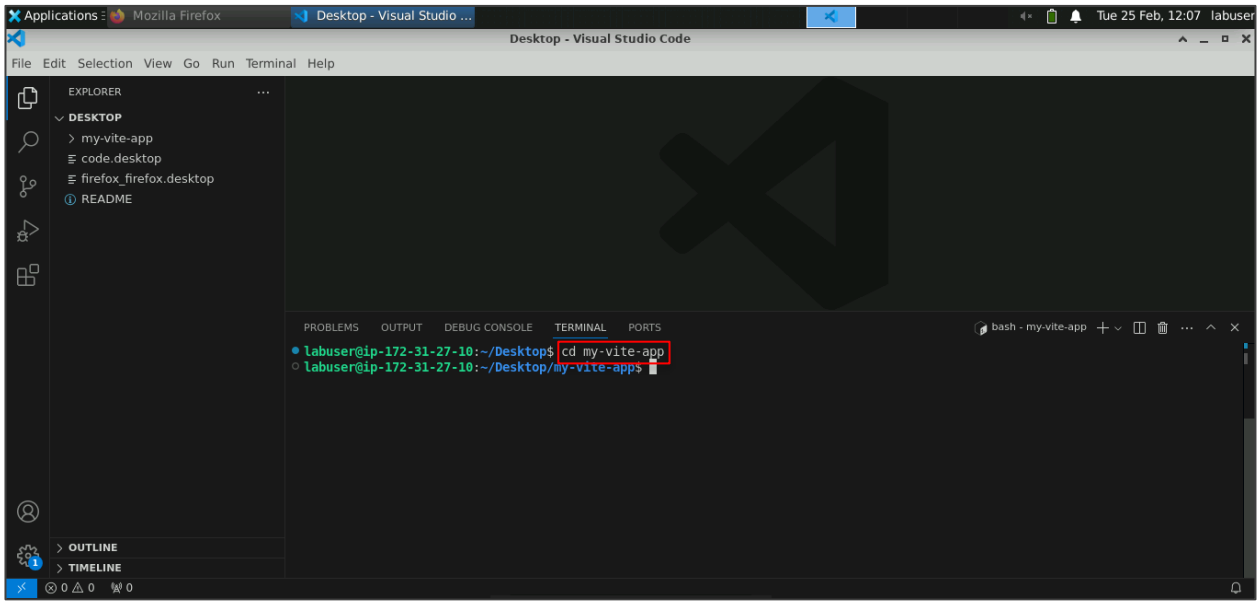


1.4 Select **JavaScript** as the language and press enter



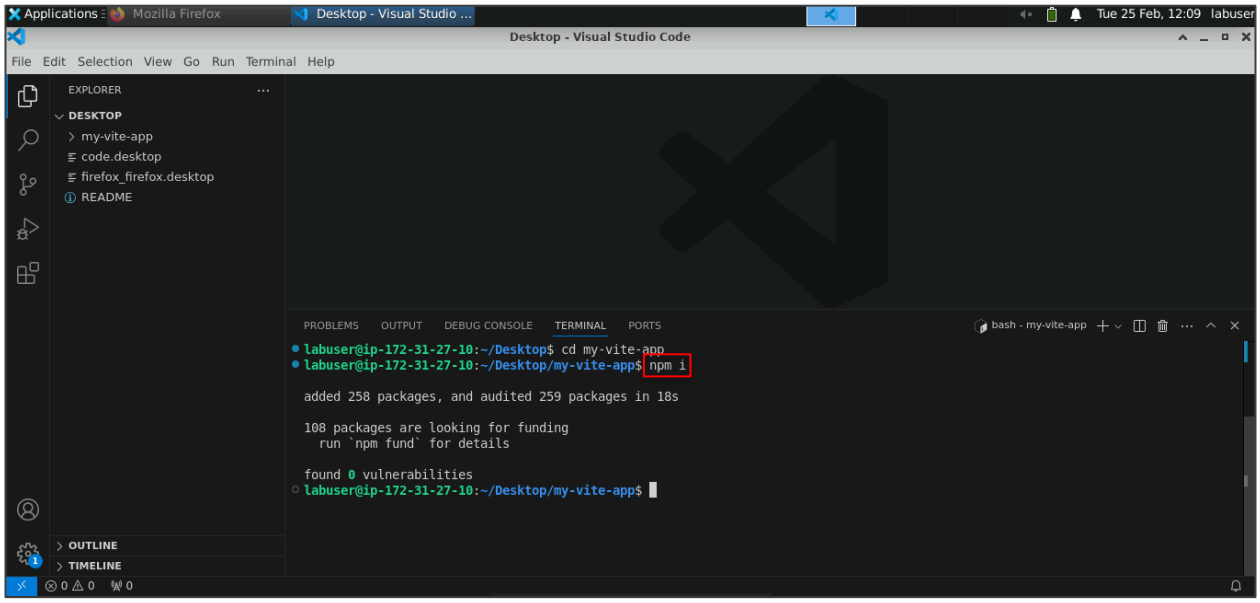
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



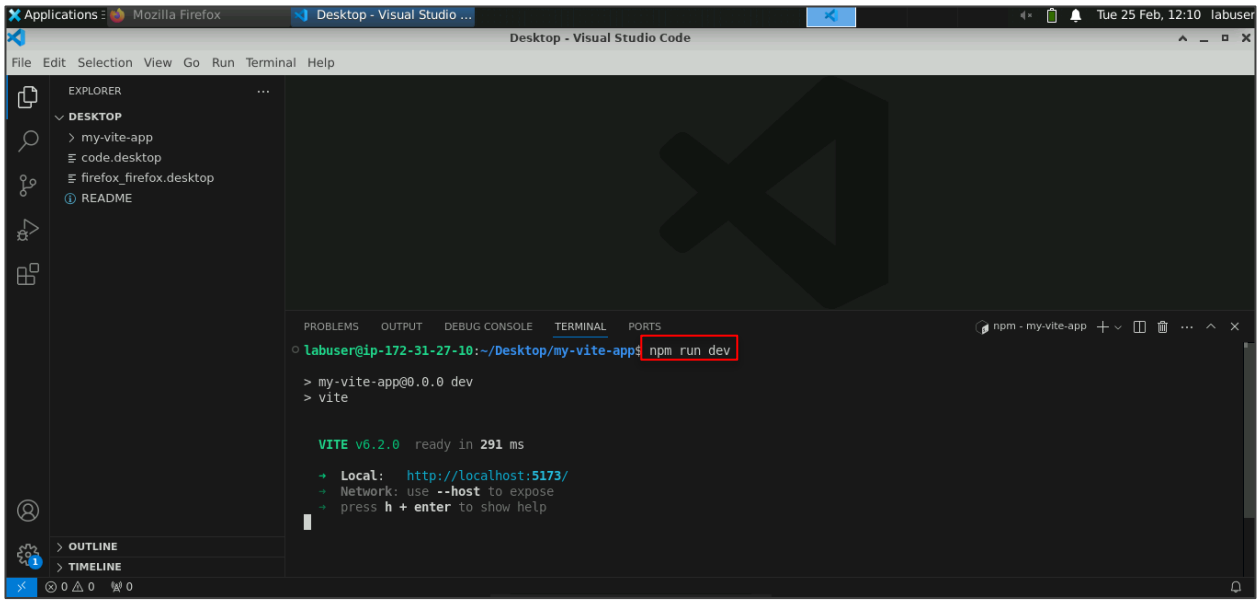
1.6 Run the following command to install dependencies:

npm i



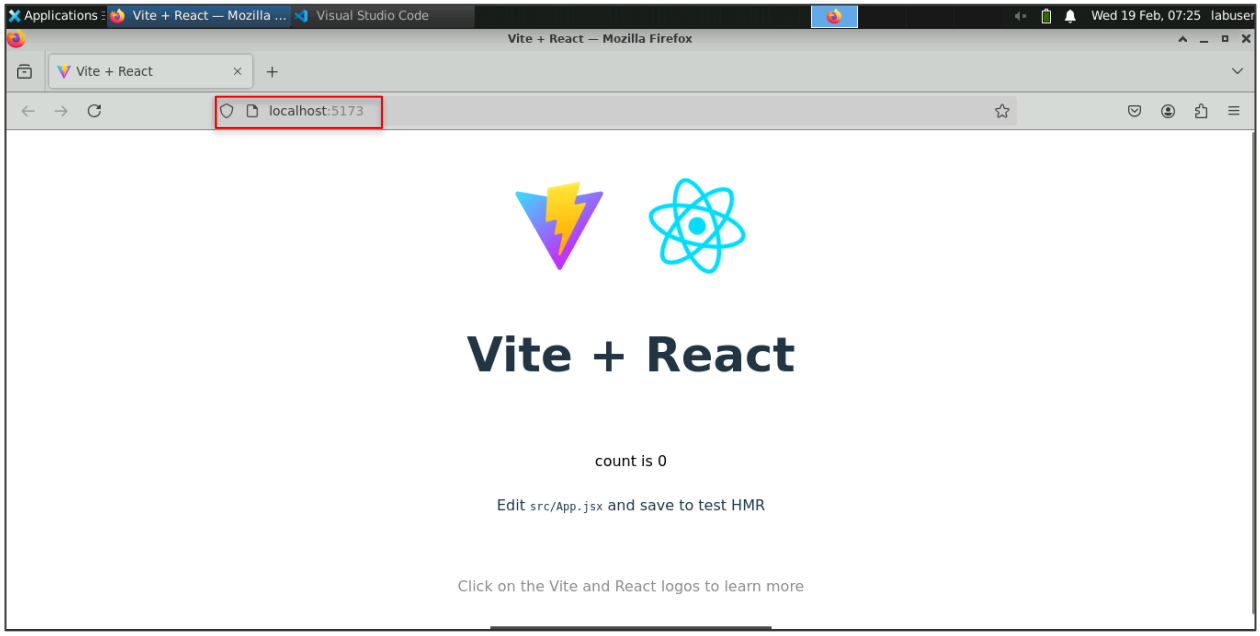
1.7 Run the following command to start the React development server:

npm run dev



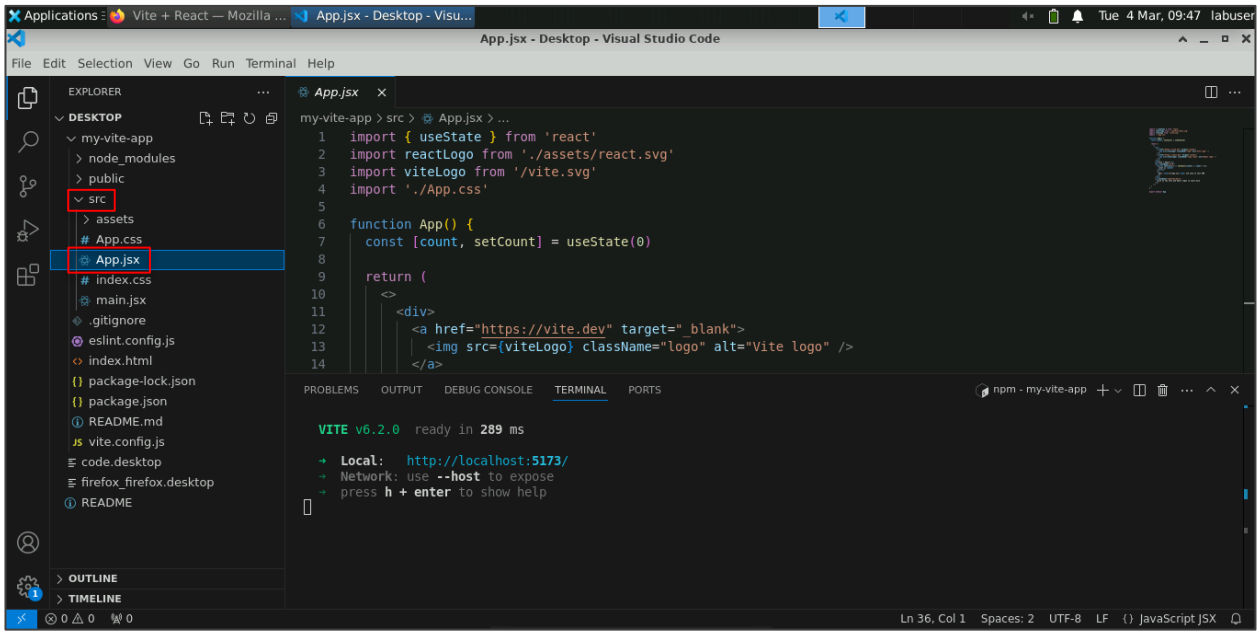
1.8 Navigate to the web browser and paste the following link:

http://localhost:5173



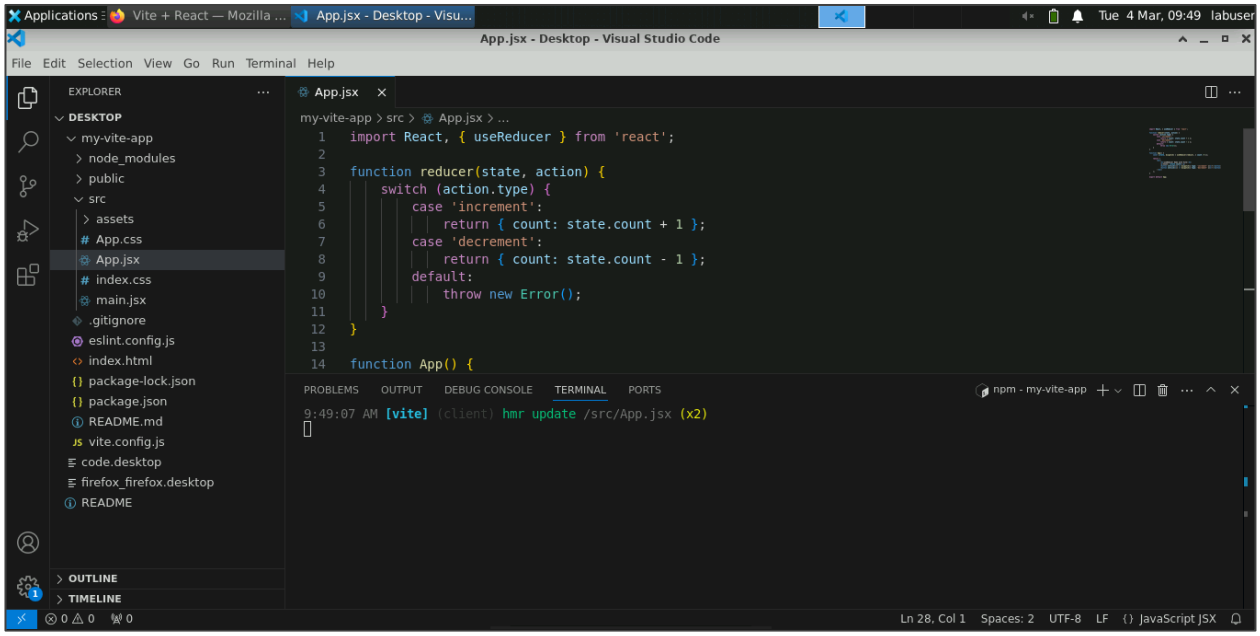
Step 2: Modify the App.jsx file

2.1 Open the **App.jsx** file inside the **src** folder



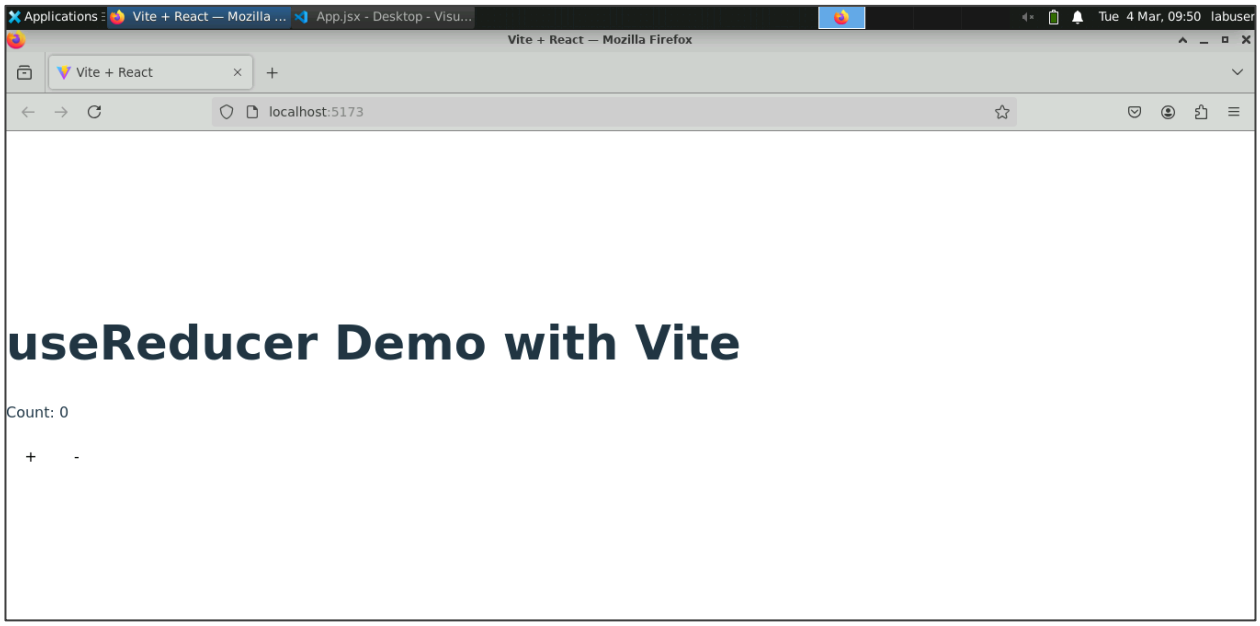
2.2 Paste the following code to include the new components in **App.jsx**:

```
import React, { useReducer } from 'react';

function reducer(state, action) {
  switch (action.type) {
    case 'increment':
      return { count: state.count + 1 };
    case 'decrement':
      return { count: state.count - 1 };
    default:
      throw new Error();
  }
}

function App() {
  const [state, dispatch] = useReducer(reducer, { count: 0 });

  return (
    <div>
      <h1>useReducer Demo with Vite</h1>
      <p>Count: {state.count}</p>
      <button onClick={() => dispatch({ type: 'increment' })}>+</button>
      <button onClick={() => dispatch({ type: 'decrement' })}>-</button>
    </div>
  );
}

export default App;
```

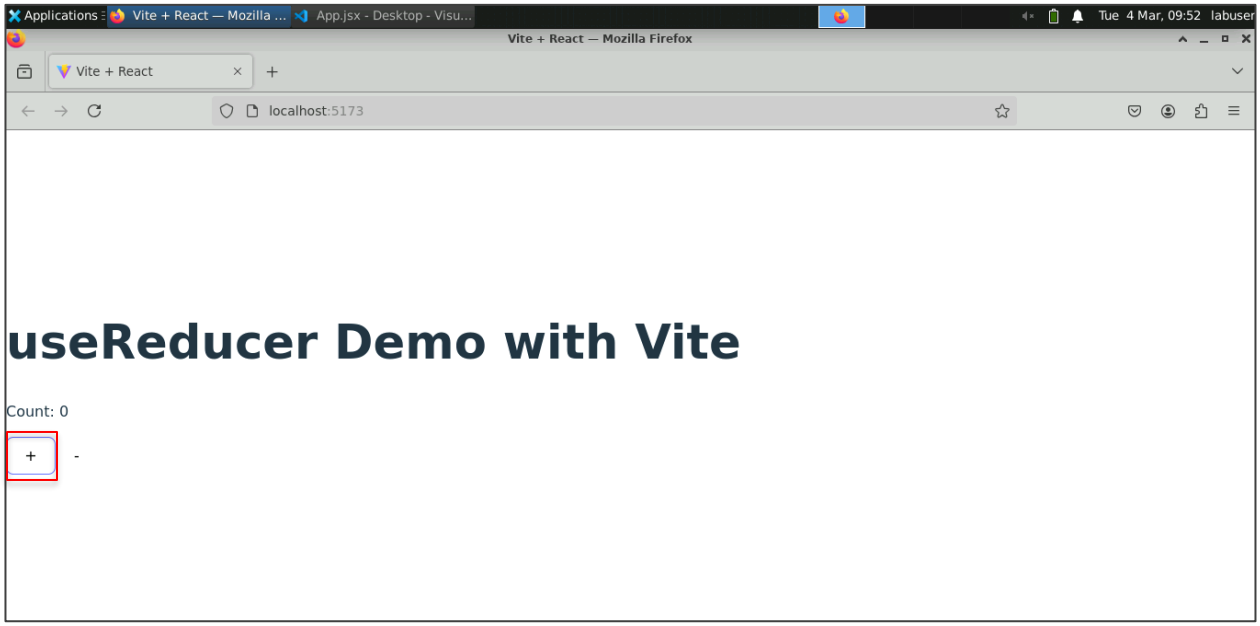


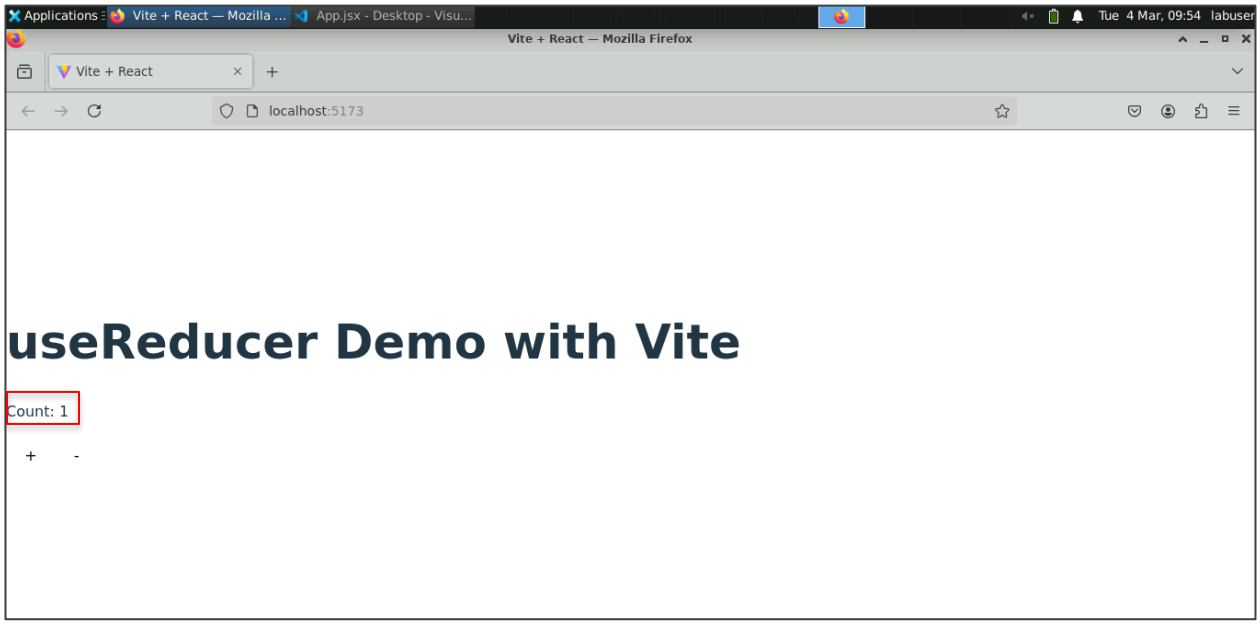
Step 3: Run and verify the application

3.1 Open the web browser and refresh the page

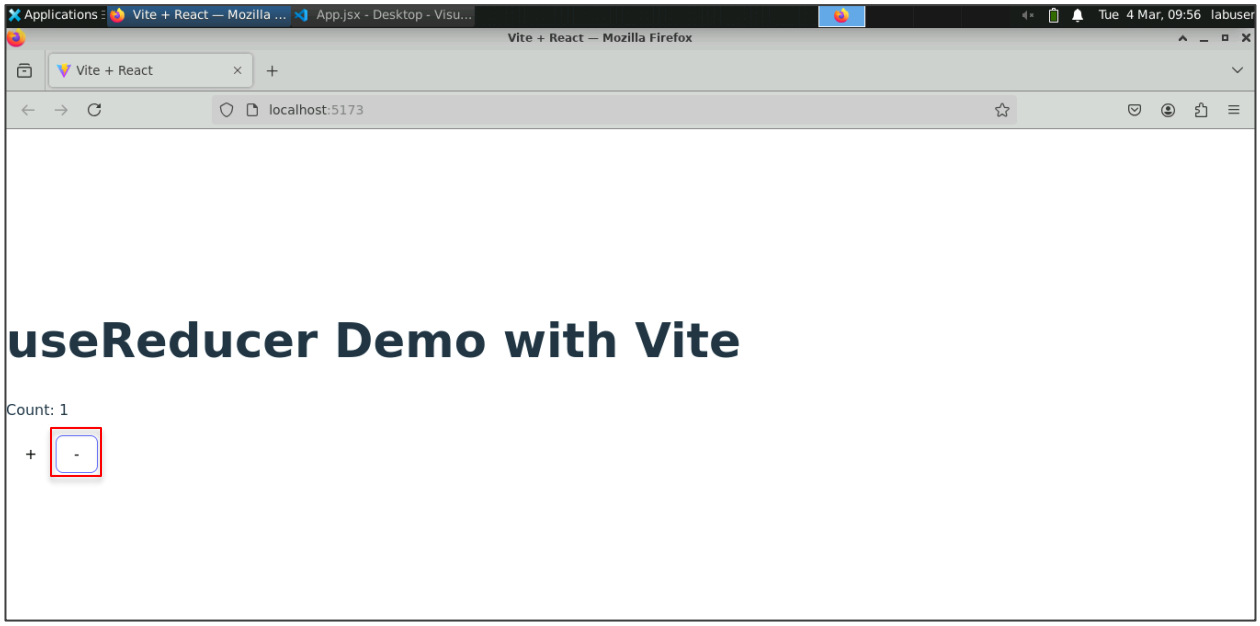


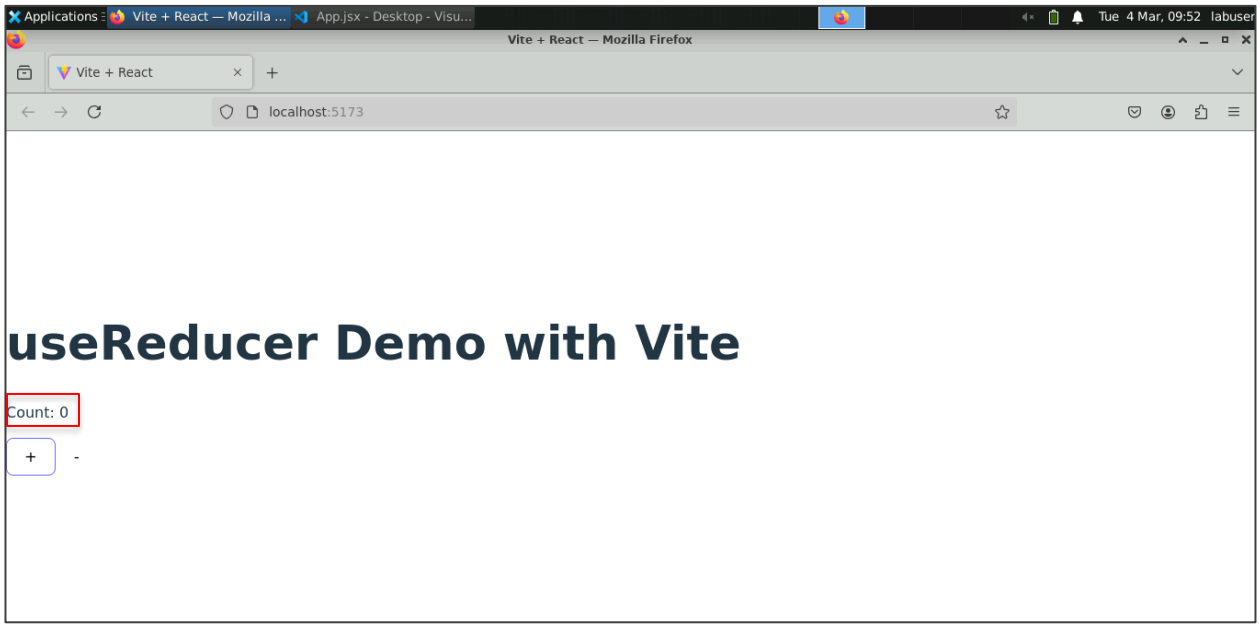
3.2 Click the **+** button to increase the count by 1





3.3 Click the – button to decrease the count by 1





By following the above steps, you have successfully created a React app with Vite and implemented the useReducer Hook for efficient state management in a scalable counter application.